STUDY MATERIAL LEC.1.1.

SUMMARY AND TRANSLATION OF CHAPTER 1, FROM THE BOOK "BIOLOGÍA CELULAR BIOMÉDICA", INTRODUCCIÓN A LA BIOLOGÍA DE LAS CÉLULAS

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Learning objectives:

- **To know the general characteristics of eukaryotic cells.**
- **To define the specific features of prokaryotic cells, comparing them to those of eukaryotic ones.**
- **To know the basic organization of subcellular agents such as viruses.**

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INTRODUCTION

Cell Biology approaches the study of cells from a comprehensive perspective including morphological, biochemical, genetic and functional aspects. Indeed, this science originates from the confluence of many other disciplines, each with its very own research methodology and approach, thereby contributing to our current multifaceted knowledge of the cell.

Cell research has been historically linked to the invention and development of microscopy, which took off in the 16th century. In the 17th century, **Robert Hooke** (1635-1703) adopted the word "cell" upon the observation under the microscope of thin sheets of cork wood which happened to be structured in a cell-based pattern, later published in his treaty *Micrographia* (1665). **Antony Van Leeuwenhoek** (1632-1723) meanwhile developed new microscopes, simpler and more precise, with the aid of which he was able to describe blood cells, protozoa, spermatozoa and bacteria, all of which he named "animalculi". Between 1837 and 1839 **Schleiden and Schwann** defined the cell theory, establishing that all living organisms are formed by cells, which in turn act as the building blocks (i.e. the structural, functional and reproductive units) of all living things. In 1885, pathologist Rudolph Virchow completed this doctrine by stating that "all cells originate from other cells".

Further improvements in microscopy and sample staining techniques allowed for the microscopic description of the different body tissues. Some remarkable contributions were made by **Camilo Golgi** (1843-1926) and **Santiago Ramon y Cajal** (1852-1934), who extended the cell theory to the nervous system by employing silver, gold and osmium salts in their studies, granting both physicians the Nobel Prize in Physiology or Medicine in 1906.

In 1930 the invention of electron microscopy constituted a major achievement, as it enabled the observation of cell organelles for the first time. Soon after, different techniques in cell fractioning and differential centrifugation appeared, allowing for the isolation of such organelles according to their sedimentation rate. Functional study of cells was made possible by these and notably by the implementation of cell culture techniques from 1940 onwards. In 1953, **Watson and Crick** proposed a double-stranded structure for DNA based on crystallographic studies. This discovery gave birth to modern genetics including the establishment of the molecular bases of genetic diseases.

In the second half of the 20th century biological research grew exponentially: many scientific achievements were made and the development of plenty of techniques resulted in a deeper understanding of the cellular structure and function. A major milestone was the Human Genome Project which undertook the sequencing of a whole human genome and stretched from the late years of the 20th century to the publication of its results in 2003. This titanic challenge was completed in a record time owing to new and powerful sequencing and bioinformatic tools. Data gathered from our entire genome has since enabled a fine yet widespread study of genetic aberrations underlying many human diseases, in addition to the progressive unravelling of gene regulation governing cell functioning.

CELLULAR ORGANIZATION OF DIFFERENT ORGANISMS

Living organisms are characterized by their ability to grow, respond to stimuli, interact with their environment, evolve and reproduce autonomously thanks to a complex cellular organization and metabolism. Although they can be divided into unicellular (bacteria, yeast, protozoa and certain algae) or multicellular (plants, animals and some algae and fungi), living organisms can be essentially classified into two types regarding their cellular organization: prokaryotes and eukaryotes. Even if the subject of our study is the human eukaryotic cell, it may be helpful to know the basic organization schema of prokaryotic organisms as well as that of other eukaryotes equally responsible for infections in humans. We will also analyze the organization of subcellular organisms (i.e. virus) equally involved in important human diseases.

Organization of eukaryotic cells

The basic inner structure of eukaryotic cells is rather conserved among all different organisms. Their main feature is the presence of a **nucleus**, which constitutes the cell's core component since it

contains the genetic material: DNA stores the necessary information to synthetize proteins necessary to the normal cell's structure and function. Unlike prokaryotes, which contain a single circular chromosome, eukaryotic nuclei contain multiple linear chromosomes, each of which can be duplicated, detached and packed away from their other copy (i.e. their sister chromatid) during cellular division. Animal eukaryotic cells possess different **cellular organelles, a cytoskeleton, 80S ribosomes and centrosomes**, and carry out internal trafficking of **vesicles** (e.g. endocytosis and exocytosis). Cellular division takes place through complex mechanisms such as those involving the nucleus (i.e. mitosis, or meiosis in the case of germ cells). As for plant cells, they are surrounded by a cellulose wall and contain chloroplasts (where photosynthesis occurs) and big vacuoles, but lack centrosomes and lysosomes. Figure 1-1 shows a scheme of the human eukaryotic cell along with an indication for each depicted structure to this book's corresponding chapter.

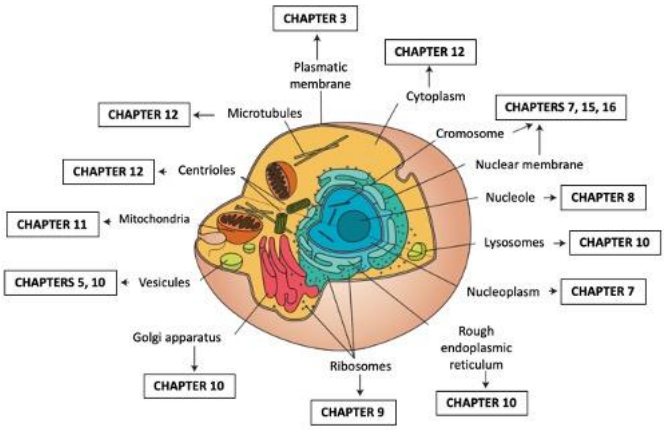


FIGURE 1-1. Overview of eukaryotic cell organization. Note: for every depicted structure the corresponding book chapter is indicated.

Next we will briefly cover the biological organization of protozoa and fungi, eukaryotes with a pathogenic potential for humans. Knowledge about their structure and infecting mechanisms is of utmost importance in order to develop effective therapies against them.

PROTOZOA

The term "protozoon" refers to a family of unicellular organisms with an eukaryotic structural organization comprising thousands of species. Most protozoic species reproduce asexually through binary fission, while others are able to carry out sexual reproduction. Some are parasitic and can infect blood cells, and the urogenital and digestive tracts among other tissues. As a general example, we will briefly present the structure and life cycle of the genus ***Plasmodium*** (fig. 1-2), which includes a group of protozoa that causes **malaria** in humans (a disease responsible for hundreds of thousands of fatalities every year, notably in tropical areas). ***P. falciparum*** is the species most frequently involved and causes the most severe form of malaria. Transmission to humans occurs through the bite of mosquito of the genus *Anopheles*, whose digestive tract serves as sexual reproducing ground for the plasmodia. From there they migrate to the salivary glands and are injected into human bloodstream following a mosquito bite. Plasmodia at this stage install in the liver where they undergo asexual division. These cellular parasites remain inside the hepatocytes until they are released back into the bloodstream, which in turn invade red blood cells, divide and successively infect other red blood cells. Inside each infected erythrocyte, they nourish from hemoglobin and replicate at some point causing cell lysis. High fever peaks and shivering typically associated to the disease are indeed due to toxic products issued from hemolysis.

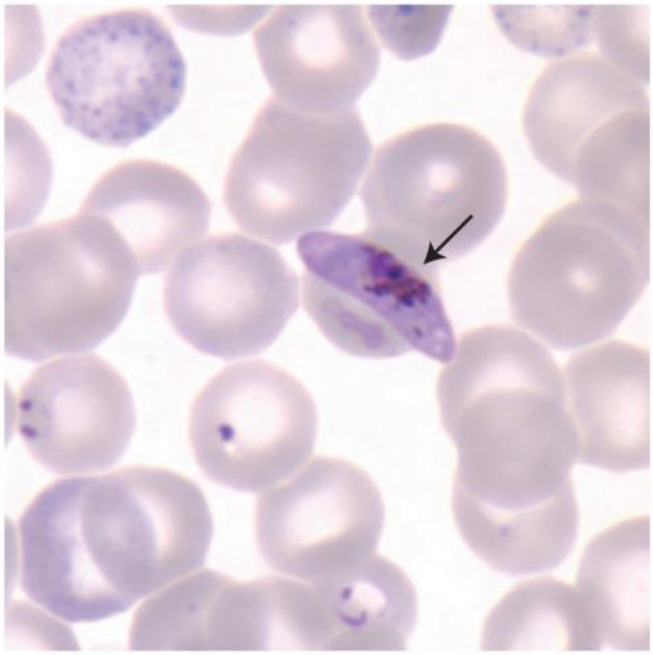


FIGURE 1-2. Blood smear micrograph depicting several erythrocytes infected by a plasmodium (arrow) (from: *http://dpdp.cdc.gov*).

FUNGI

Fungi are a group of eukaryotes with heterotrophic metabolism comprising both **unicellular and pluricellular** species with **sexual and asexual** reproduction. Unicellular fungi that are rounded-shaped are known as **yeasts**. Most other fungi are pluricellular and are formed by long cylindric cells assembled in rows to constitute filaments known as **hyphae**. These can grow to form macroscopic structures or **mycelia**. Their organization is typically eukaryotic with a surrounding **wall of polysaccharides** as distinctive feature. This wall constitutes a rather stiff exoskeleton with a protective function. Disease caused by fungi are known as **mycosis**. Certain species are fairly relevant from a clinical perspective since they behave as opportunistic pathogens in immunosuppressed patients (e.g. *Candida sp.* and *Aspergillus sp.*).

Candida albicans is a yeast that can be found as part of the commensal microbiota of the skin and mucosae of healthy individuals. When ambient conditions are appropriate (i.e. warm temperature, high humidity and decreased/impaired immune response) it can trigger disease with varying symptomatology. **Mucocutaneous candidiasis** most commonly affects the hands, feet, genitalia, axilla and buccopharyngeal cavity, and involves erythematous and pruritic lesions, papules and pustules. In case of severe immunodepression the yeast is able to migrate into the bloodstream (a condition known as **candidemia**) and affect organs such as the eyes (risk of blindness) and the kidneys (risk of renal failure, decreased urinary output). Systemic manifestations may include intravascular coagulation, fever and shock, and may well be fatal.

Some filamentous fungi may be responsible for cutaneous, subcutaneous or systemic mycosis. The genus ***Aspergillus*** (fig. 1-3) is constituted by opportunistic species responsible for systemic infections, most frequently at a pulmonary level. It may produce asthma, bronchitis, otitis and even **fungaemia** of very poor prognostic.



FIGURE 1-3. Micrograph of *Aspergillus sp.* hyphae.

Another notable species of fungus is ***Saccharomyces cerevisiae***, a yeast used in the food industry to make bread and beer, but also in biological research as a widely studied model. As a matter of fact, these yeasts can be easily grown in culture and have long been used in laboratories for the study of multiple aspects related to DNA replication, gene expression, cell division and the life cycle.

Organization of prokaryotic cells

Prokaryotes are unicellular organisms that reproduce **asexually and lack nucleus** and, normally, membrane-bound organelles. They are organisms distributed ubiquitously in nature, which relates to their metabolic diversity, proof of their extraordinary degree of adaptation to their environment. Table 1-1 presents the differences in structure and function between eukaryotic and prokaryotic organisms. Prokaryotes can be classified into two big groups or domains: **bacteria** (aka. eubacteria) and **archaea** (aka. archaebacteria). Representatives of the latter group are extremophilic organisms (i.e. with the capacity to survive in highly adverse environmental conditions), which lack any medical interest; so we shall focus specifically on bacteria.

TABLE 1-1. Review of the differences between eukaryotes and prokaryotes

	Eukaryotes	Prokaryotes
Cell size (mean diameter)	10-100 µm	0,2-2 µm
Genome location	Inside the nucleus, covered by a nuclear membrane	In the cytoplasm, without a surrounding nuclear membrane
Genome structure	Linear, segmented into chromosomes, bound to hystones	One single circular chromosome; without hystones
Cytoskeleton	Present	Absent
Cell wall	Absent in animal cells; cellulose in plants; quitin in some fungi; glycans and manans in veasts	Peptidoglycan and other structures
Mobility	Cillia and flagella in some animal species	Rotatory flagella for locomotion
Cell division	Mitosis or meiosis	Binary fision
Ribosomes	80S in the cytoplasm	70S
Membrane bound organelles	Nucleus, mitochondria, rough endoplasmic reticulum, Golgi apparatus, endosomes, vacuoles; chloroplasts in phosynthetic plants	None. Some bacteria have mesosomes (primitive organelles)

TABLE 1-1. Review of the differences between eukaryotes and prokaryotes.

BACTERIA

Bacteria (fig. 1-4) are prokaryotes surrounded by a **cell wall** that grants them protection against their environment. According to their wall structure, bacteria can be classified into two large groups: **gram-positive** and **gram-negative**, depending on their coloration after Gram staining. Some bacteria can be classified as **acid-alcohol-resistant**: their walls are highly hydrophobic. A classic example is ***Mycobacterium tuberculosis***, responsible for tuberculosis.

Bacteria display a great diversity of shapes and aggregations characteristic to each species and, therefore, with a certain taxonomic interest. The most common among them are **cocci** (sphere-like), **bacilli** (rod-like) and **spirilla** (spiral-like). They equally display extraordinarily diverse metabolisms: strictly aerobic, strictly anaerobic and anywhere in between; with capacity to carry out fermentation, cellular respiration, photosynthesis, and even capable of obtaining energy from the oxidation of inorganic compounds. Some are highly plastic in their behavior and will adapt to the outer conditions by using one or another system to obtain energy.

Prokaryotes **lack membrane-bound organelles** even if some have pseudo-organelles (e.g. mesosomes where organic and inorganic compounds are stored). Prokaryotes' **nucleic acids are not surrounded by a nuclear membrane** and so DNA, ribosomes, enzymes and metabolites are found distributed all throughout the cytoplasm. The typical bacterial genome consists of a double-stranded circular DNA molecule (i.e. the bacterial chromosome) not bound to histones. Bacteria can also possess several chains of extra-chromosomal circular DNA known as **plasmids**. Their sequences are much smaller than the bacterial chromosome and codify genes that confer them certain advantageous traits for survival such as antibiotic resistances, adaptations to adverse environmental conditions and virulence factors. Among other prokaryotic characteristics, it should be noted that protein translation is performed by **70S ribosomes** and that their **cell membrane lack cholesterol and steroids**.

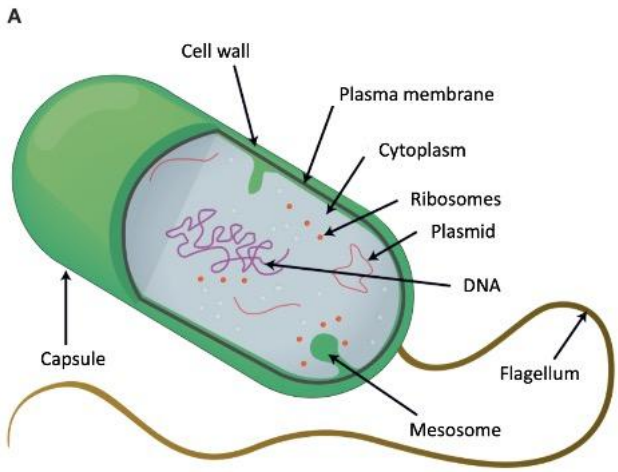


FIGURE 1-4 (A). Bacterial cellular organization schema: general organization with its main structures.

The structure and composition of the bacterial wall differ between gram-positive and gram-negative bacteria. The essential wall component is **peptidoglycan**, which provides rigidity and stabilizes its structure. The wall of gram-positive bacteria is made up of a thick peptidoglycan layer while that of gram-negative bacteria is thin, it is

separated from the plasma membrane by the periplasmic space and it is surrounded by a second membrane layer rich in lipopolysaccharides. The **lipopolysaccharide** is a potent endotoxin capable of triggering fever, the activation of the complement cascade, an increase in cytokine levels, etc. Some bacteria, whether gram-negative or gram-positive, possess yet an outer layer or **capsule** (aka. **glycocalyx**) usually made up of polysaccharides. This capsule keeps the cell from desiccation and phagocytosis, and it allows bacteria to adhere to certain substrates and tissues.

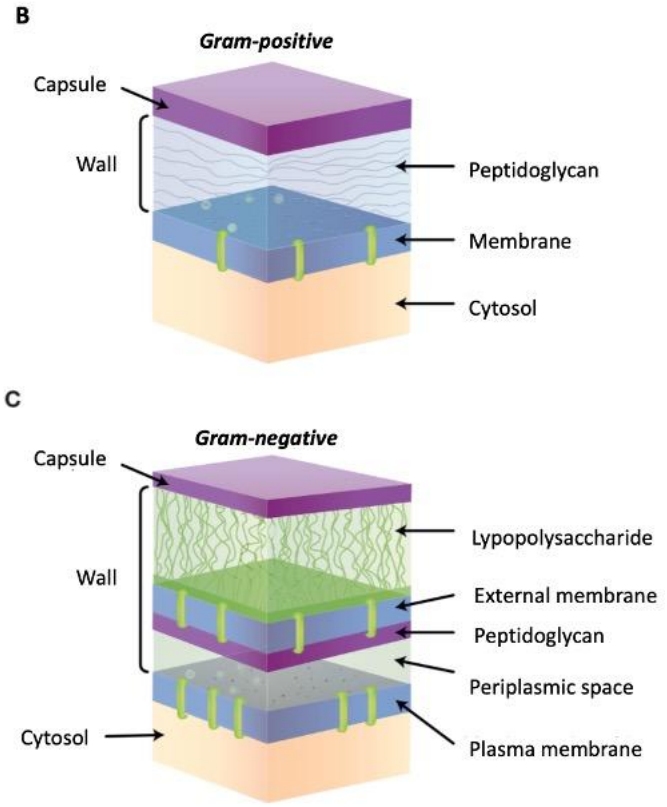


FIGURE 1-4 (B and C). Bacterial cellular organization schema: gram-positive **(B)** and gram-negative **(C)** cell wall composition.

Bacteria may also have one or several **flagella** which they use mainly for locomotion. For further adherence they may exhibit protein projections named **fimbriae**. **Pili** (pilus in singular) share a similar structure yet they are longer and thicker, and they specialize in **bacterial conjugation** that leads to the transfer of genetic material among bacteria. Some species (most of which are gram-positive bacilli) are able to form **endospores** when facing harsh environmental conditions; it constitutes indeed a fairly resistant form to chemical and physical injury. Dormant bacteria in the form of endospores can survive in such environments for hundreds of years and eventually germinate when conditions become favorable, returning to their normal state.

There are uncountable examples of pathogenic bacteria, among which we can provide few examples.

Helicobacter pylori is an acidophilic gram-negative species with a characteristic helix shape colonizing the human gastric mucosa. This bacteria adapts well to the very acidic conditions imposed by stomach secretions. *H. pylori* causes a chronic infection that can remain asymptomatic or lead to ulceration and even perforation of the gastric walls; it may also increase the risk of suffering from gastric cancer and lymphoma. In the past, when this microorganism's involvement in gastric ulceration was unknown, this pathology had a difficult management. Nowadays, an adequate combination of antibiotics and proton pump inhibitors constitutes an effective, quick and cheap treatment.

Another example we will consider is ***Vibrio cholerae***, a gram-negative curved flagellated bacillus with facultative aerobic metabolism and causal agent of **cholera** outbreaks. *V. cholerae* (among other *Vibrio* species) is transmitted through the intake of contaminated food. As with other gastrointestinal pathogenic bacteria, it evades the host's immune system (including lysozyme action, gastric acid, phagocytic cells, etc.) and adheres to the intestinal epithelium, invades the epithelial cells and alters the water-salt balance, causing, causing vomits and diarrhea.

Escherichia coli is a gram-negative bacillus with facultative anaerobic metabolism normally found in the bowel (among other organs) in **symbiosis** with its host as part of the regular intestinal microbiome. Some strains are, nonetheless, highly pathogenic and are responsible for intestinal, urinary and other sorts of infections. This bacteria has great importance in biomedical research: *E. coli* is the most widely used bacteria in genetic engineering aiming at the obtention of compounds of biologic and therapeutic interest. Many recombinant proteins are produced by genetically modified E. coli. It has also been the subject of numerous biological studies such as DNA replication and transcription. Mutant variants have allowed us to better understand key aspects of bacterial antibiotic resistance.

Organization of pathogenic sub-cellular agents

VIRUSES

Structure and replication

Viruses are **infectious agents** that consist basically in a genome enclosed by a protein envelope and lack the ability to replicate autonomously (fig. 1-5). They are obliged **intracellular parasites**, meaning they must enter a foreign cell and utilize the host's enzymes and substrates in order to replicate. Despite viruses having a great variety of shapes and structures, as well as of replication strategies and invading mechanisms, they all share some basic features. Their shell, known as **capsid**, is formed by subunits (mostly of protein origin) called capsomeres. On top of it, some viruses **can present a lipoprotein envelope**, a piece of membrane issued from the cells where they replicated. Their genomes can consist of **DNA or RNA** (not both at the same time), whether single or double-stranded, linear or circular, segmented or not.

For simplicity, we will divide viruses into three classes according to their type of nucleic acid and their replication model: 1) DNA viruses; 2) RNA viruses; and 3) viruses that require an inverse transcriptase.

1. **DNA viruses** can have double-stranded (dsDNA) or single-stranded (ssDNA) genomes. They replicate with the aid of a DNA polymerase codified either by the viral genome or by that of the host cell. Examples of dsDNA viruses are **adenoviruses**, typically responsible for airway and gastrointestinal infections.

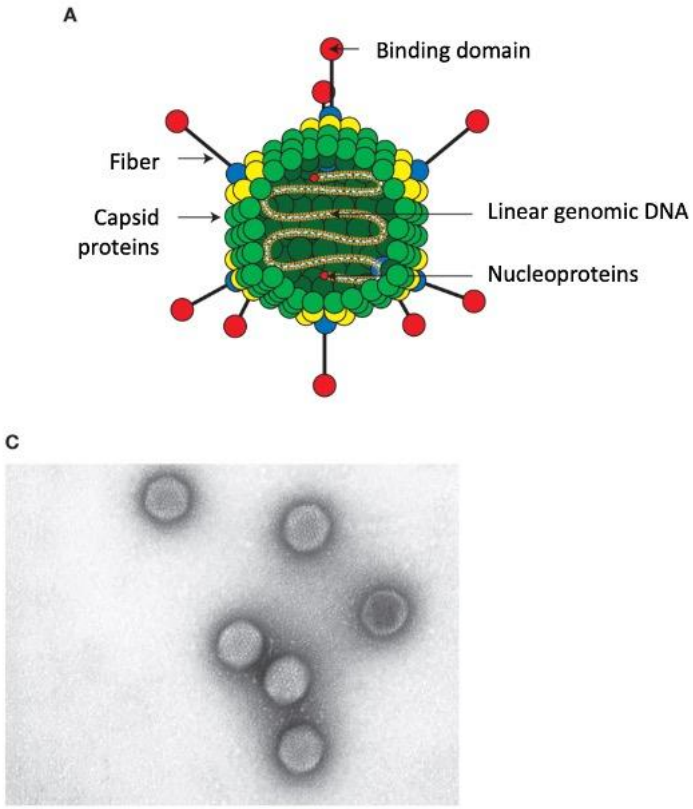


FIGURE 1-5 (A and C). Structure of some viruses: basic structure **(A)** and high-power electron micrograph **(C)** of adenoviruses (from *http://www.med.upenn.edu/gtp/morphology_gallery.shtml*).

2. **RNA viruses** can equally be subdivided for having dsRNA and ssRNA genomes. They must produce a RNA-dependent RNA polymerase (known as replicase) for replication.
3. A third type includes viruses that codify for an **inverse transcriptase** that synthesizes DNA from a RNA template. **Retroviruses** are a group of viruses that include HIV (responsible for AIDS).

Viral capsids are typically icosahedral (i.e. with 20 faces, each of which display an equilateral triangular shape) or cylindric (formed by proteins with a helicoidal layout). Some bacteriophage (i.e. viruses infecting bacteria) capsids can be more complex (fig. 1-5). Once translated by the host's enzymatic machinery, capsomeres self-assemble around the viral genome. Some viruses additionally possess a lipid envelope issued from the infected cell and so it contains not only viral proteins (fundamental for facilitation of future cellular invasion) but also components of the host cell.

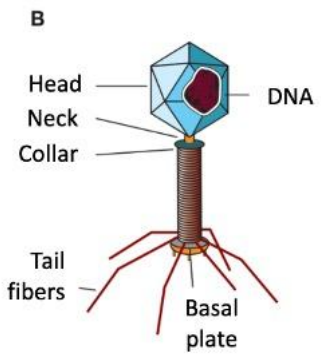


FIGURE 1-5 (B). Structure of some viruses: basic structure of a bacteriophage.

Among the numerous viruses responsible for human diseases, we will briefly cover the human papillomavirus, AIDS virus and SARS-COV2 (COVID-19 virus.

Human papillomavirus

Human papilloma virus (HPV) is a membrane-less virus with **icosahedral capsid and a circular dsDNA genome**. The infectious viral particles do not codify their own polymerase; rather, replication of the viral genome is carried out by the host cell's polymerase (fig. 1-6). After DNA entrance into the cell, RNA transcription takes place, followed by translation of viral proteins and the generation of new virions (i.e. complete viral particles ready to infect other cells). Viral DNA remains inside the nucleus in the form of **episomes** (i.e. without integrating the host's genome) and replicates during the division phase of the host cell, leading to chronification of the disease. Generally speaking, HPV infects the squamous epithelium (e.g. the skin's keratinized epidermis and the genital mucosa). A common sign of HPV infection is the presence of warts or papillomas essentially composed of squamous cells growing as benign tumors. HPV transmission typically occurs through direct contact with the skin of genitalia. Although these warts are benign, the integration of the viral genome into that of the host cell is associated to malignant progression.

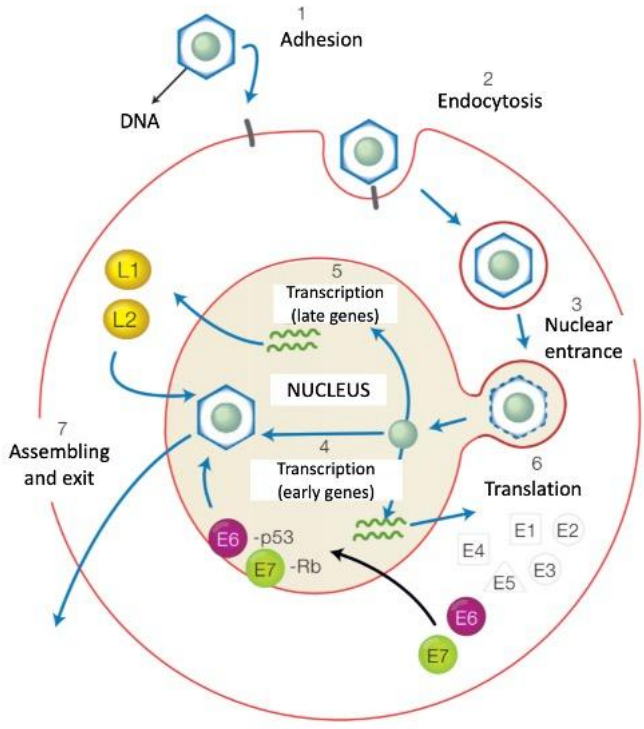


FIGURE 1-6. Human papillomavirus (HPV) infectious cycle schema. Note: this is an oncogenic virus with the potential to cause cervical tumors, among others.

Immunosuppressed individuals infected by HPV present a high risk of developing carcinomas. Even though there exist numerous types of papillomavirus, each variant usually displays a very specific tropism. While some infect the skin and can give rise to plantar

warts, others have a genital tropism. HPV-16 and HPV-18 are strongly associated to genital cancer. The high prevalence of HPV infections among sexually active women and the increased risk they have to develop cervical cancer have fostered the development of vaccines that effectively prevent the viral disease.

Human immunodeficiency virus

The human immunodeficiency virus (HIV) (fig. 1-7) infects cells that express **CD4** on their surface, like **T helper lymphocytic cells and macrophages**. The infection propagation and the death of these cells lead to **acquired immune deficiency syndrome** (AIDS), which increases the patients susceptibility to opportunistic infections. HIV is a retrovirus, so it belongs to the group of viruses which code for an inverse transcriptase. Within the capsid, the virus contains three enzymes: a **reverse transcriptase**, a **protease** and an **integrase**. The capsid is surrounded by viral matrix proteins associated to an enveloping membrane.

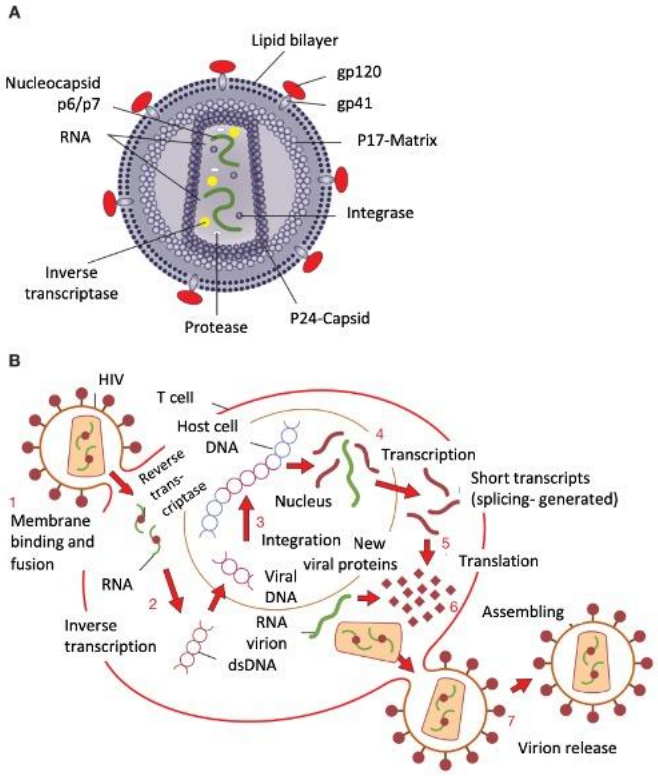


FIGURE 1-7. HIV virus structure **(A)** and infection cycle **(B)** schemas. Note: the basic infecting mechanism is explained in the text below.

Union of the virus to CD4 allows the entrance into the cell through pore formation and fusion with the plasma membrane (fig. 1-7, B,). Once inside, viral replication requires the action of the inverse transcriptase: viral RNA is transcribed in the cytoplasm to form a DNA copy (fig. 1-7, B). The DNA chain serves as template for the synthesis of pro-viral dsDNA, which is transported into the nucleus where it integrates the host cell's genome with the help of the viral integrase. The provirus can remain in dormancy (and so the patients stay asymptomatic), although in the presence of certain transcription factors the host cell's RNA polymerase will transcribe viral mRNA to be translated in the cytoplasm (fig. 1-7, B). The assembling and release of new viral particles takes place at the plasma membrane of the host cell (fig. 1-7, B).

Coronaviruses and SARS-COV2 (COVID19)

Coronaviruses are responsible for 30% of common seasonal colds. However, some variants can become very virulent for the human species, such as SARS-Cov2, originated in China in December 2019, causing the disease called COVID19 (coronavirus disease, 2019). This type of virus has a **zoonotic origin**, since it probably migrated to humans from bats. Structurally, it is an **RNA virus** surrounded by an envelope. Inserted in this outermost layer are a series of proteins, among which the **S (spike) glycoproteins** stand out, which give it the corona appearance that characterizes this family of viruses. It is precisely the S protein that the virus uses to interact with its future host cell through a membrane-bound receptor of the host cell (fig. 1-8). Alveolar cells of the lungs are the most likely to be infected.

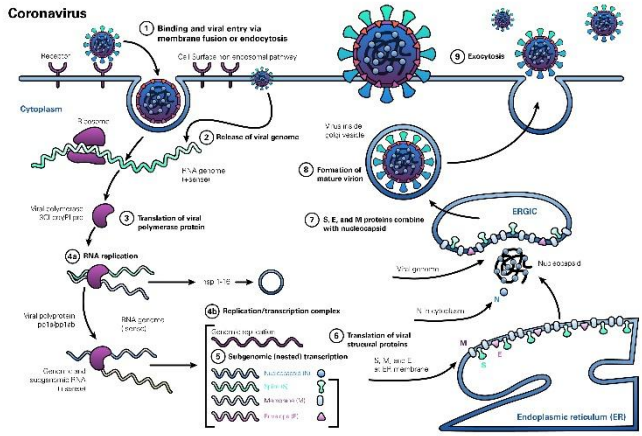


FIGURE 1-8. Infectious and replicative cycle of SARS-COV2 (COVID19).

Once inside the cell, the viral RNA gets released into the cytoplasm. This is followed by the process of translation of viral proteins and proteases and the replication of the viral RNA itself (fig. 1-8).

The structural proteins of the future virion envelope migrate and insert into membranes of the endoplasmic reticulum-Golgi apparatus. In turn, the viral RNA binds to proteins to form the nucleocapsid, which will migrate to vesicles of the Golgi apparatus where it will assemble together with the other components to form the mature virion. Finally, mature virions are released to the outside by exocytosis (fig. 1-8).